



Name	Id number	Program
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BIT1123 OBJECT ORIENTED PROGRAMMING
ASSIGNMENT 1 - INDIVIDUAL
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ALL SECTIONS
FACULTY OF INFORMATION TECHNOLOGY
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1. Introduction

This reflective report is a summary of my learning experience in Object Oriented Programming tutorials in Java. This is an opportunity to consider what knowledge and programming skills I have gained in the tutorials and to explain what I found difficult and how I overcame it.

The tutorials used various Java programs and helped me to get a better understanding of the concepts of object-oriented programming. Additionally, I learned how to carry out programming work effectively with the help of GitHub and how to keep my repository of tutorials organized and structured.

The java source code is in my GitHub repository and contains programs from various tutorial weeks; programs with students, persons, lecturers, employees, appliances, refrigerators, washing machines and a graphical quiz application. This gave me some idea about the applicability of programming concepts with various examples and exercises.

2. Knowledge Gained from Each Tutorial

Week 1

Week 1 was beginning to use simple java classes and objects. I found Main.java and Student.java files in the repository which were useful for me to understand how the main program and the class interact.

I studied how to write classes, attributes and to use objects within a java program. This gave an insight into the concepts of Object Oriented Programming.

Week 2

During week 2, I continued with the practical tutorial exercises which helped me understand java programming better. The tutorial activities helped me get to know the syntax of Java and the structure of Java programs.

I was also able to get more comfortable with creating and editing source-code files and structuring my programming activities.

Week 3

In Week 3, I collaborated with multiple classes (Person, Student, Lecturer, and Main) and a Main class.

This tutorial gave me the concept of different classes representing different objects and for the relationships between classes, it gave me the idea of how to do that in an object oriented program. This also helped with an understanding of class design and object creation.

Week 4

Week 4 continued the development of my Java and object-oriented programming skills. The practical exercises enabled me to use concepts from different tutorials learnt and get more comfortable to work with more than one java class.

The ability of using separate source files also had me understand the significance of structuring the code into well-defined classes.

Week 5

For Week 5, there are Student.java and main.java in the repository. This tutorial provided me with more practice with java classes and objects.

Was able to write classes and utilize from the main program. Through practice, I learned the value of having readable, well-organized Java code.

Week 6

The 6th Week includes employee.java, lecturer.java and main.java. These examples helped me to get a clearer idea of a relation of different classes.

In the process of using examples of employees/lecturers, it was easier to grasp how similar concepts can be conveyed through different classes. It also helped me to get more comfortable with object-oriented class design.

Week 7

Main.java, the refrigerator classes Refrigerator.java, the washing machine classes WashingMachine.java, and the Appliance.java class are in Week 7.

It was helpful to understanding how to use object-oriented relationships. The example with the appliances shows that various kinds of appliances can be represented as separate classes.

The classes for refrigerator and washing machine enabled me to gain an understanding of the organisation of the classes, as well as further understanding of reusable structures in object-oriented design.

Week 8–9

There are 3 files in the folder Week 8-9: Main.java, Main.class and task.txt.

The activities provided another hands-on experience with using Java programming. I got more comfortable reading programming tasks, implementing solutions and testing my code.

I also found that the viewability and manageability of the programming work improved when the source files and the supporting materials are kept together.

Week 10

The code for week 10 is in Questions.java and QuizBattleGUI.java.

The tutorial was helpful because it introduced a more practical application of java programming with a quizzing application using a graphical user interface.

I learned that Java can be used to make a GUI program with QuizBattleGUI.java. This boosted my confidence with the application of programming concepts to a more complete application.

3. Challenges Encountered

There were some difficulties in the tutorials which was to understand how different Java classes interact with each other. In some programs, there may be several classes and it might be hard to see which class contains which data or functionality.

Another difficulty was the concepts of the object-oriented paradigm and how to use them in a proper way when creating real-world programs. Understanding the concepts was easier after repeating the implementation through various examples.

One of the difficulties I had in writing and testing Java code was that little syntax or class or method or program structure mistakes can cause the code to fail to run as expected.

The other challenge was to structure my programming work into a repository on GitHub. Naming and organization in folders were important to keep the files for the tutorials separate.

4. How the challenges were overcome

I worked through these problems primarily by making lots of practice and by reading the teaching materials. I did not try to grasp every concept in one go, but worked through each example step-by-step.

If there were any errors, I would ensure that the Java classes have the correct structure and then read the code through carefully to see if there were any errors. Repeating the program made me learn how various changes in the program would lead to various outputs.

Using several examples, I also understood that there were some patterns in the java language. For instance, after implementing classes for instance Student, Person, Lecturer, and Employee, I felt more at ease with how classes can be used to represent real world objects.

I also updated my GitHub organization, organizing all of the tutorial work under separate folders and having one repository for each subject.

5. Improvements in My Java Programming Skills

Gradually had my java programming skills improved throughout the tutorials. Initially, I started with the basics of the structure and syntax of a java program.

I practiced creating classes, objects, attributes, and methods after doing additional tutorials, and started to feel more comfortable with them. I also got a bit more comfortable using several Java source files within one program.

With the help of the practical examples containing students, the lecturer, employees and appliances, I was able to make sense of the fact that Java can be used to model real world objects.

I also got a better idea of how to use the Java programming language for a more complete application, not just simple console-based exercises with Java, in the Week 10 quiz application.

I have in general, become more confident in my programming skills and more proficient in structuring Java projects, through the tutorials.

6. Understanding of Object-Oriented Programming Concepts

The tutorials helped me to gain more knowledge about Object Oriented Programming.

The concept of classes and objects was one of the important ones I learned. A class is a blueprint, an object is an instance of a class.

I also gained a better understanding of the ways that classes could be useful to represent real-world entities. For instance, there are classes in the repository, for example, Student, Person, Lecturer, Employee, and Appliance.

The examples with the Refrigerator and WashingMachine also helped me to understand how related classes can be developed in an object oriented system.

In general, the tutorials made me realize that OOP enables programs to be structured in a way that makes them easy to understand and reusable.

7. Future Learning Plans

I would like to improve my java skills through working on larger projects and more practical application in the future.

I want to develop my skills in advanced OOP (Object Oriented Programming) concepts and understand inheritance, polymorphism, encapsulation, abstraction, interfaces, exception handling and more.

Also, to enhance my GUI and to make complete Java programs would be great.

Furthermore, I will continue to use Git and GitHub with my programming projects as I believe version control and project organization are skills of software development.

8. Conclusion

All in all, the Object-Oriented Programming tutorials helped me get some hands-on experience in Java. I learned how to create and use classes and objects, and increasing my ease in structuring Java programs into several source files.

I could use programming concepts in various situations by using the different tutorials presented such as students, persons, lecturers, employees, appliances and the quiz application.

I have also gained and improved programming problem solving skills and how to take my work and organize it on GitHub.

I had some problems in the tutorials, but overcoming these problems helped me to get a better understanding and feel more confident in programming with Java. I feel that the lessons I learned in these tutorials will help a lot in making future advanced Java and software development skills.

9. githup link

https://github.com/dhoomity963-code/binmahmood-abdulrahman-sabri-a-omar-_op